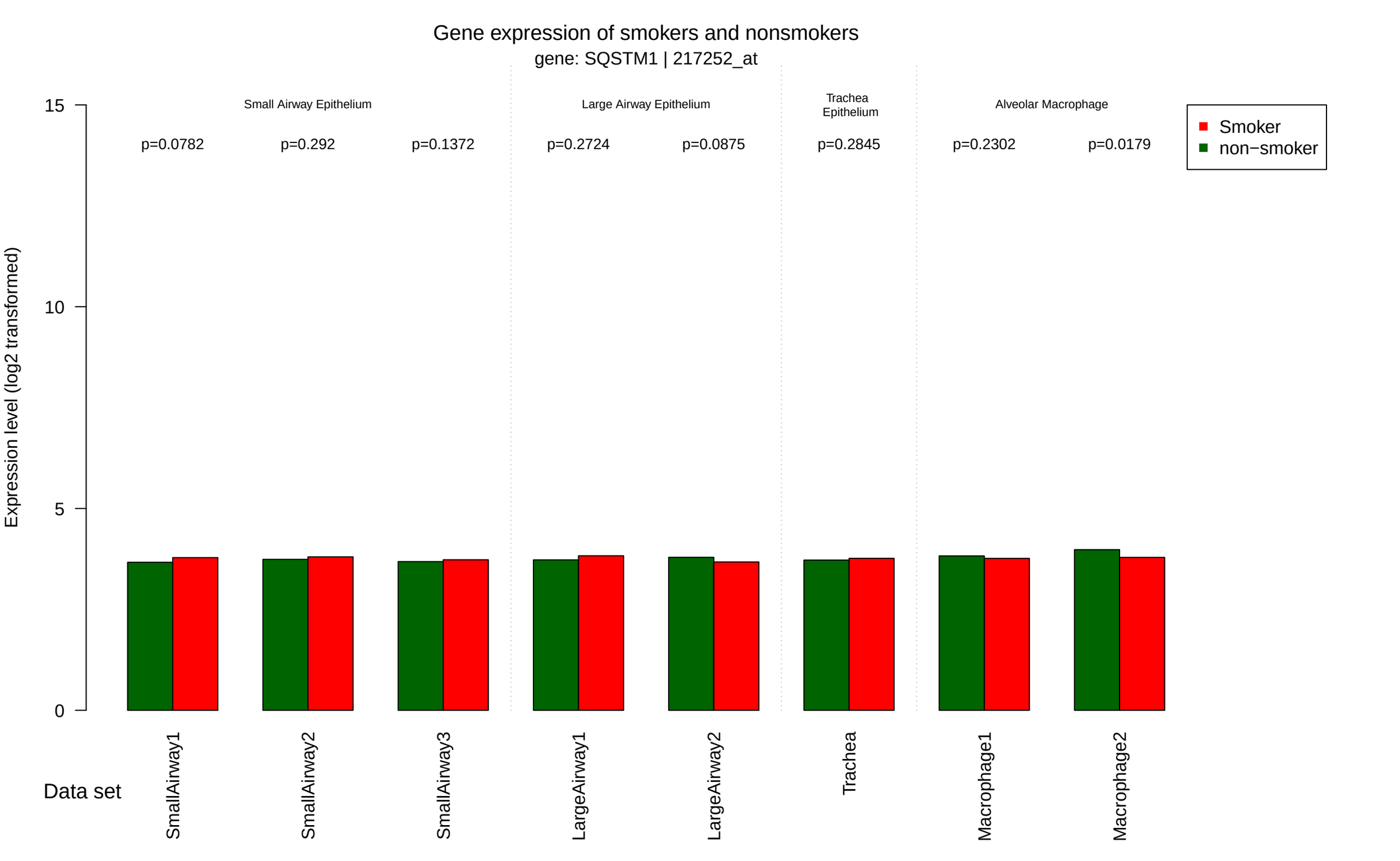
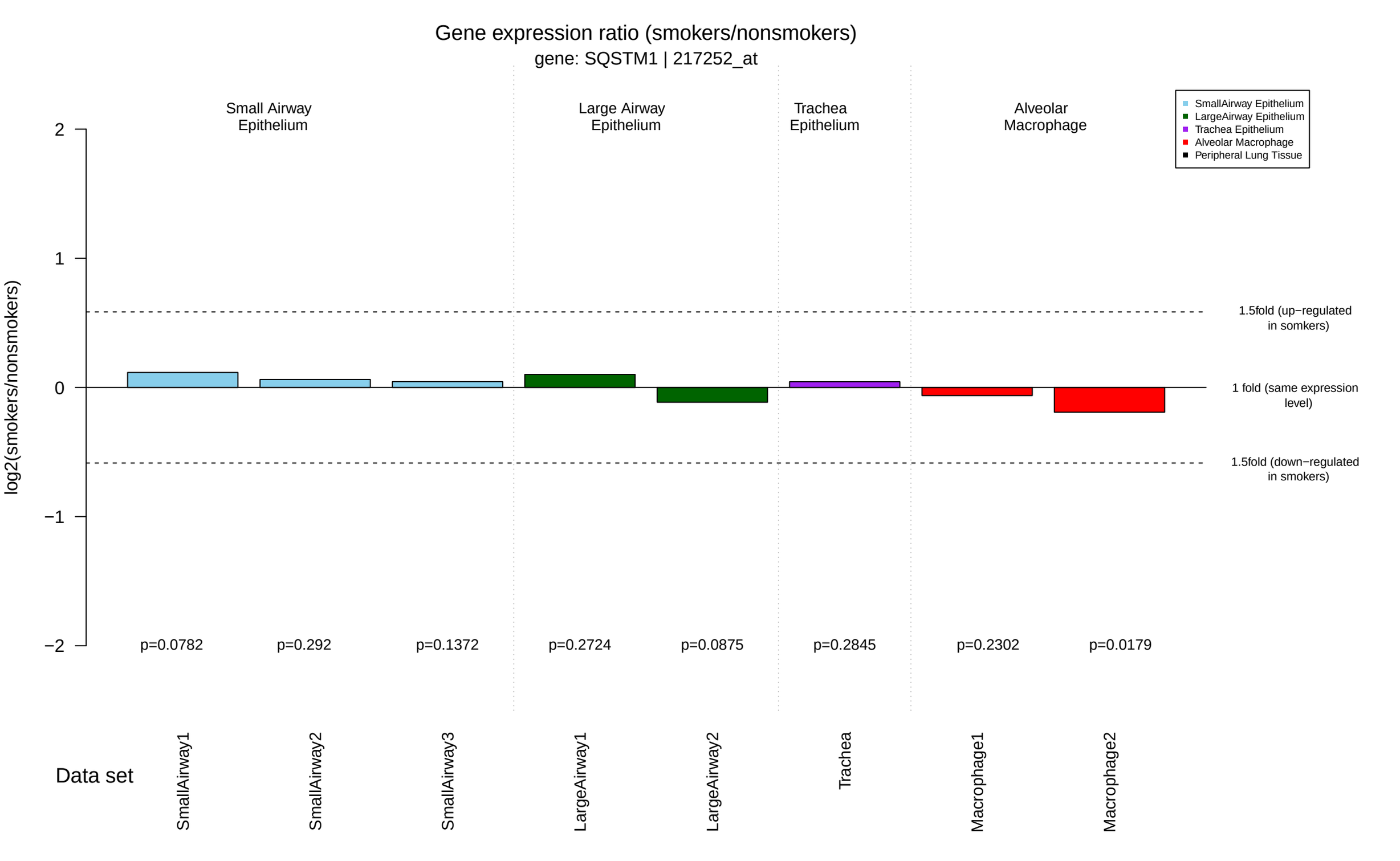
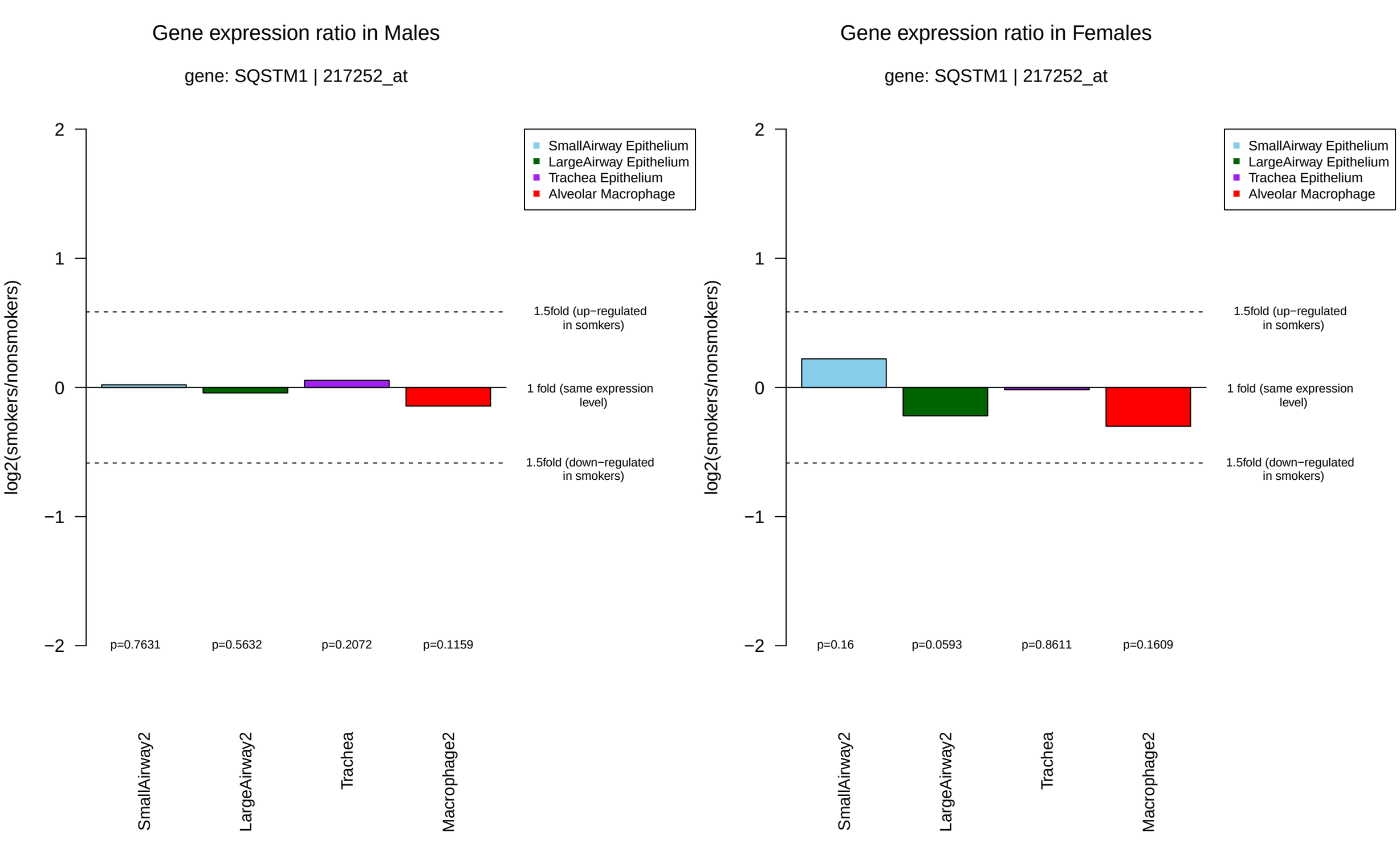
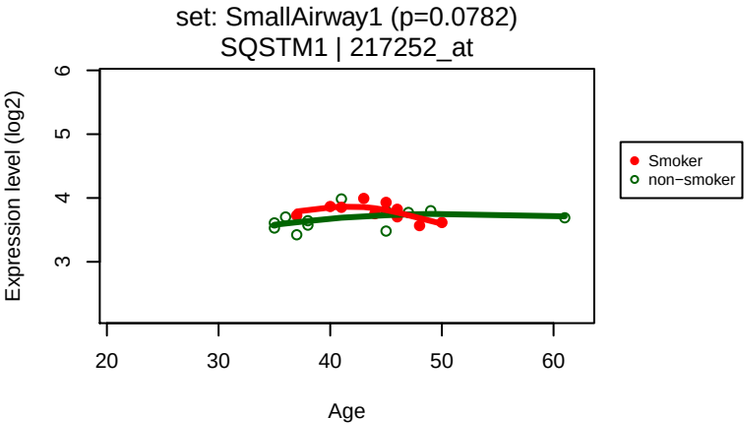
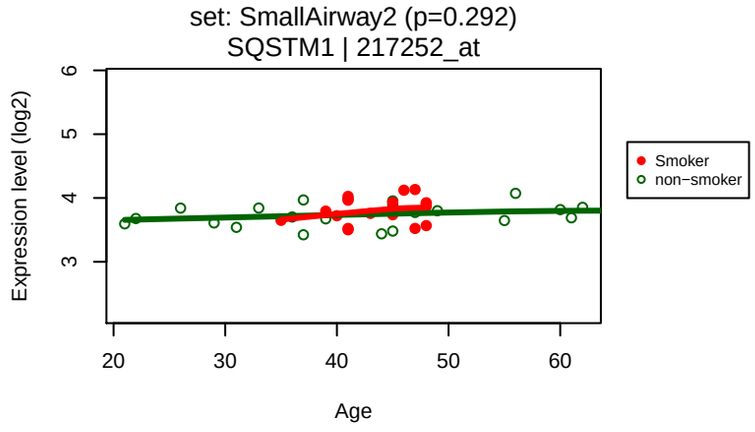
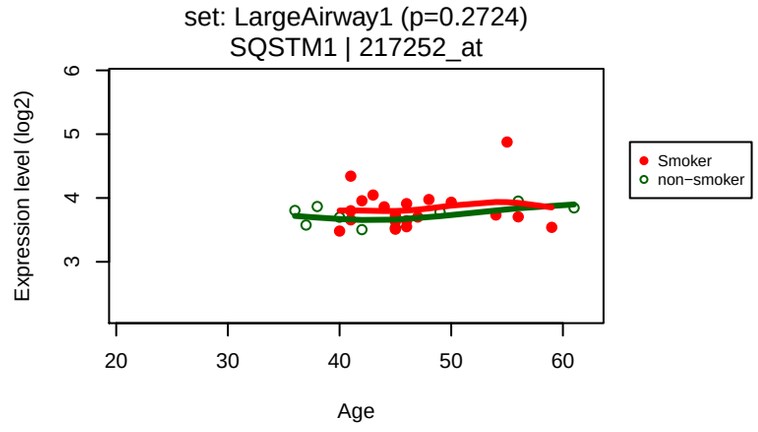

Gene expression of smokers and nonsmokers
gene: SQSTM1 | 217252_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0782
p=0.292
p=0.1372
p=0.2724
p=0.0875
p=0.2845
p=0.2302
p=0.0179
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

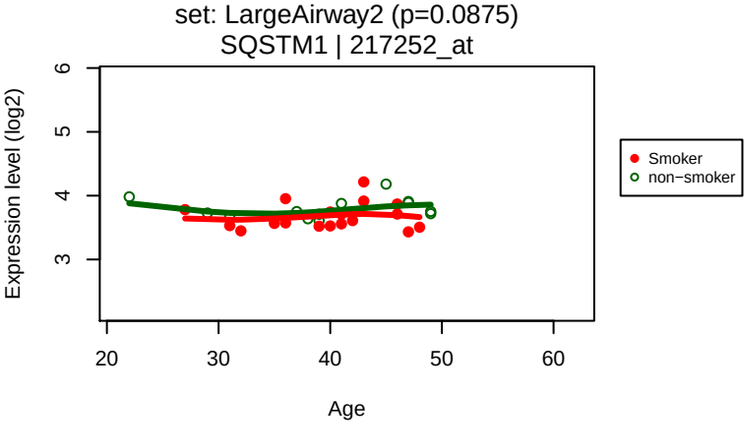

Gene expression ratio (smokers/nonsmokers)
gene: SQSTM1 | 217252_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0782
p=0.292
p=0.1372
p=0.2724
p=0.0875
p=0.2845
p=0.2302
p=0.0179
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

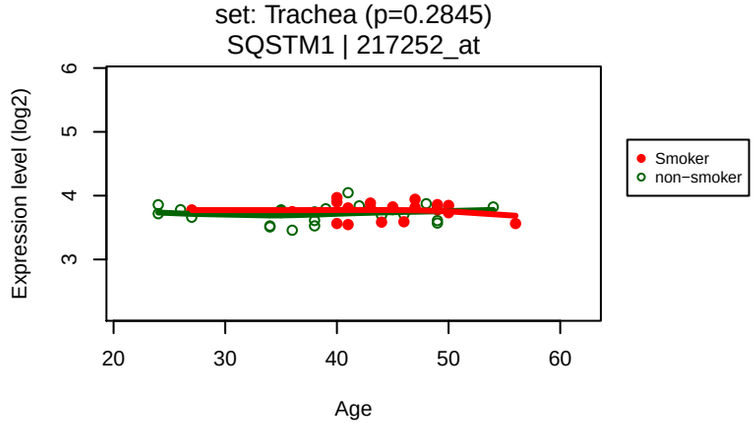

Gene expression ratio in Males
gene: SQSTM1 | 217252_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7631
p=0.5632
p=0.2072
p=0.1159
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SQSTM1 | 217252_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.16
p=0.0593
p=0.8611
p=0.1609
SmallAirway2
LargeAirway2
Trachea
Macrophage2

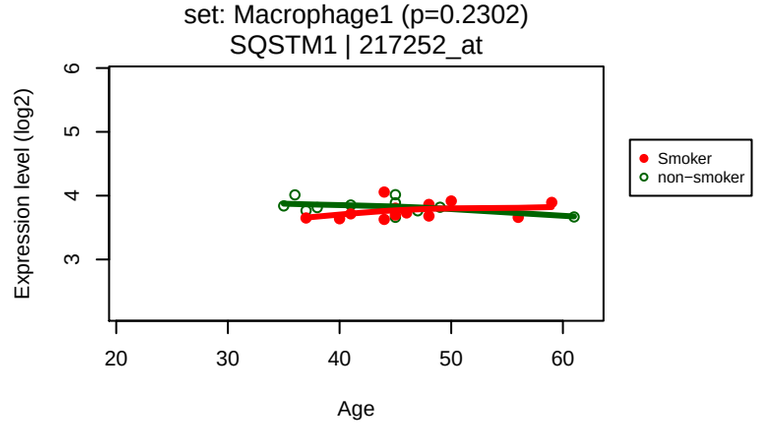

set: SmallAirway1 (p=0.0782)
SQSTM1 | 217252_at
Expression level (log2)
Age
Smoker
non−smoker

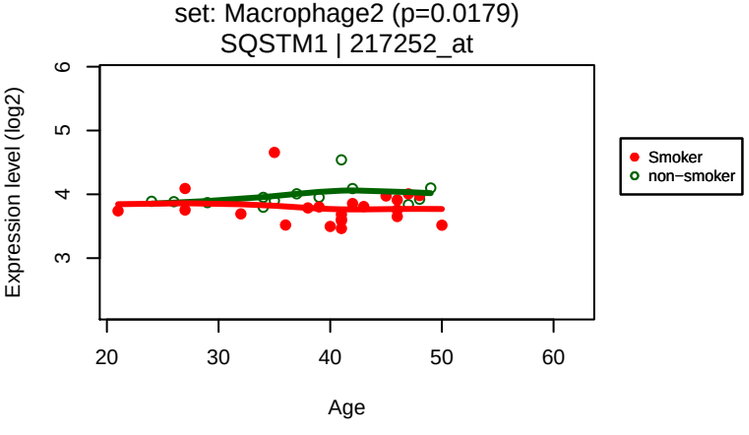

set: SmallAirway2 (p=0.292)
SQSTM1 | 217252_at
Expression level (log2)
Age
Smoker
non−smoker

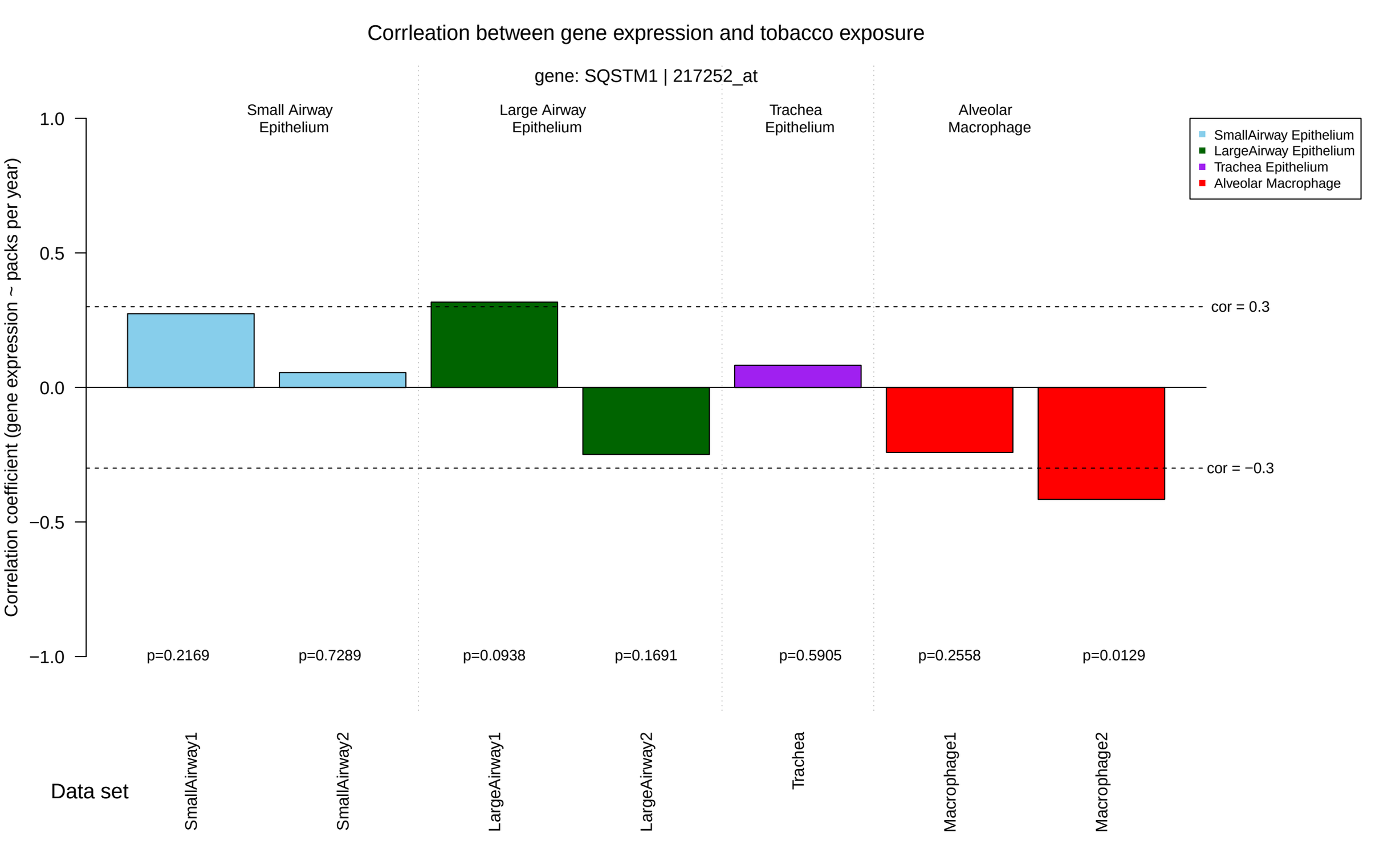

set: LargeAirway1 (p=0.2724)
SQSTM1 | 217252_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0875)
SQSTM1 | 217252_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2845)
SQSTM1 | 217252_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2302)
SQSTM1 | 217252_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0179)
SQSTM1 | 217252_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SQSTM1 | 217252_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2169
p=0.7289
p=0.0938
p=0.1691
p=0.5905
p=0.2558
p=0.0129
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2